



Synthesis of triazole library

Automated synthesis using ISYNTH workflow in Chemspeed FLEX Isynth

PharmBioTec's department of Drug Discovery focuses on the synthesis of small, drug-like molecules for a broad range of therapeutic targets. Usually, many compounds are needed to identify active species, or to improve the activity/pharmacokinetic profile of an initial hit. Instead of employing time and labor-intensive bench work, PharmBioTec has chosen another path, that of automation. Towards this end PharmBioTec has acquired the most flexible platform for organic synthesis on the market: Chemspeed's ISYNTH.

This modular robotic platform enables library synthesis and work-up in easy-to-use disposable reactors. It is suited for low and high temperatures, for handling liquids / solids and working under inert or reactive gas pressure atmosphere.

Workflow



Objectives

- Synthesize a library of 32 triazoles in a fully automated protocol, including work-up and sampling, starting from commercially available boronic acids and alkynes
- Compare yields obtained in the manual and automated procedures

Experimental Set-up

1. ISYNTH, a software-driven robotic platform containing:
2. Solid Dispensing Unit
3. 4-Channel Liquid Handling Unit
4. 20 mL Disposable Glass Reactors
5. Phase-separation cartridges for automated work-up

Results

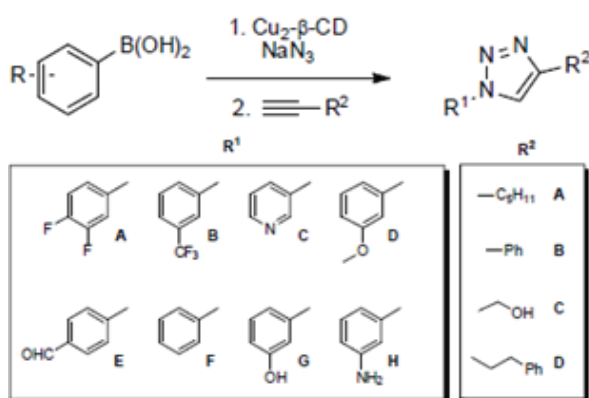
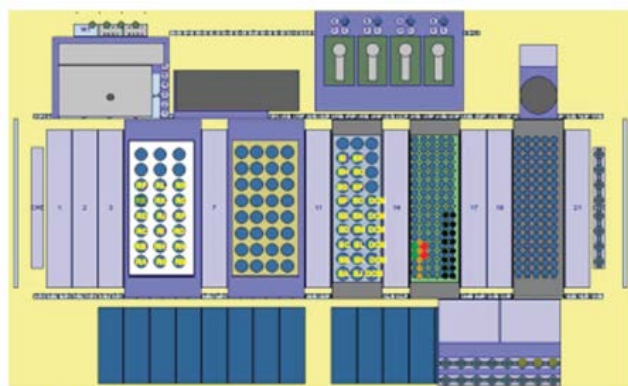


Figure 1: Overview of prepared compounds



A: RA-RR (Reaction-Zone)
B: EA-ER (Extraction-Zone)
C: DCM
Cu-CD-Complex, Sodium Azide, Alkynes, HPLC-Samples, MgSO₄

Figure 2: Schematic layout of the experiment, showing storage positions of raw materials and reaction wells.

All reaction steps, including work-up, were performed in the ISYNTH platform.

Crude products were purified by preparative HPLC if their purity was below 95%. The average yields obtained were above 30% and comparable to manual procedure.

Conclusion

A library of 32 triazoles was synthesized on the ISYNTH platform, with an average purity of 96% (after HPLC). Generation of such libraries is achieved at the same time as it takes for a single compound manually.

